

Communicating with Data – Welcome!

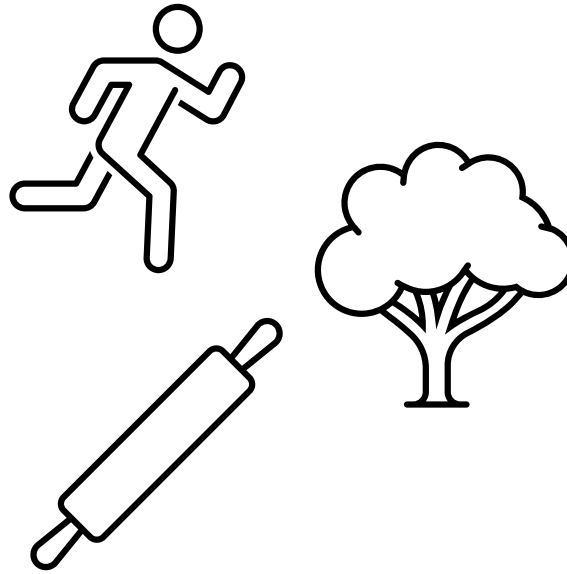
Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

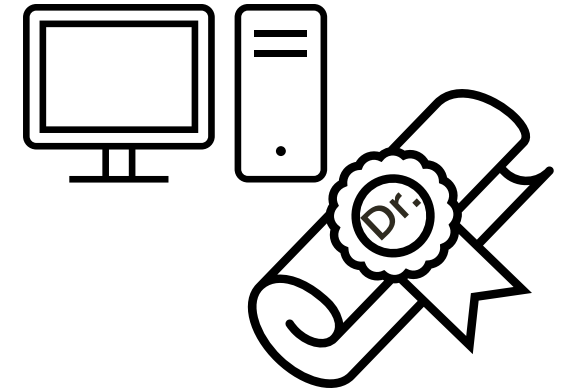
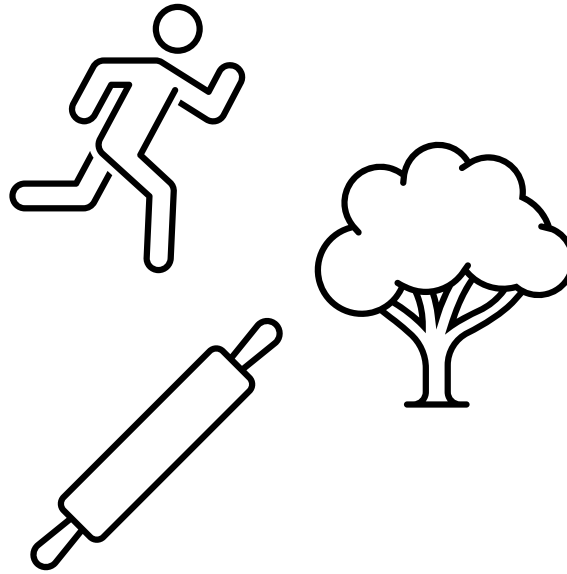
Plan for Today

- Intros
- About this course
- How is visualization useful?
- What is data?
- Structure of this course

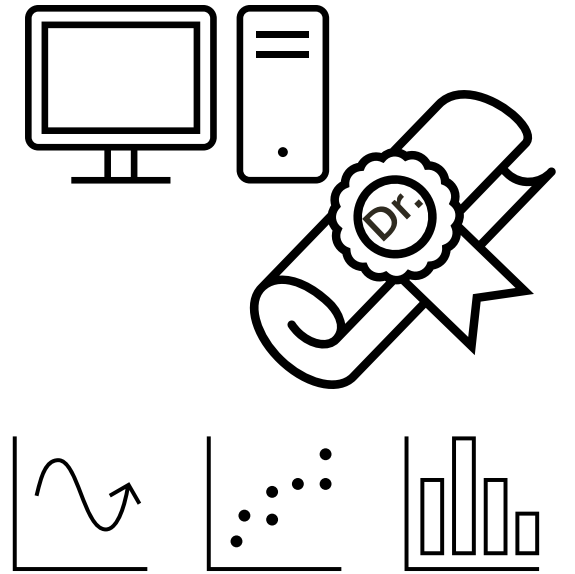
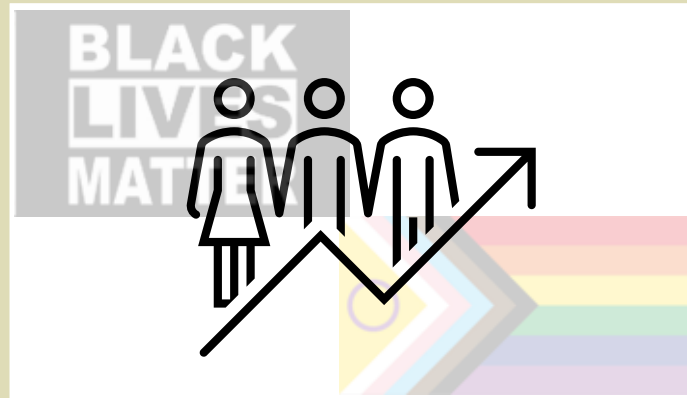
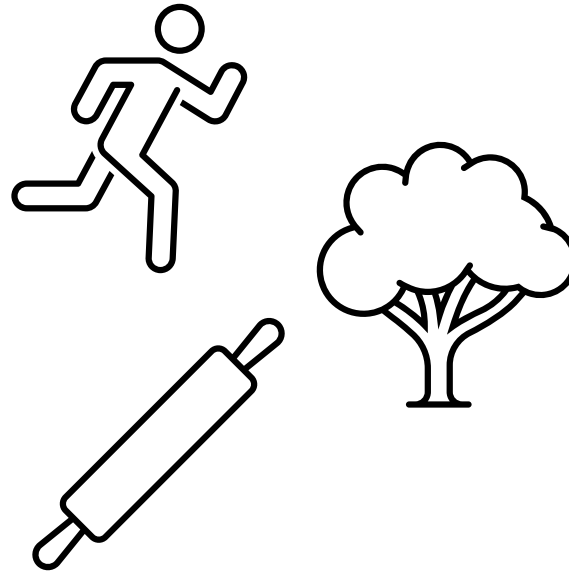
Who Am I?



Who Am I?



Who Am I?



Who Are You?

- Form groups of 3
- Introduce yourselves (name, pronouns)
- Share:
 - A highlight of your summer break
- Find 1 thing that your entire group has in common (favorite color? hometown? left-handed? Be creative!)
- After about 5 minutes we will go around, introduce ourselves, and share what each group has in common

Who Are You?

- Form **new groups** of 3 (move around!)
- Introduce yourselves (name, pronouns)
- Debate:
 - Is a hot dog a sandwich?
- Be prepared to share your stance

Nametags and Attendance

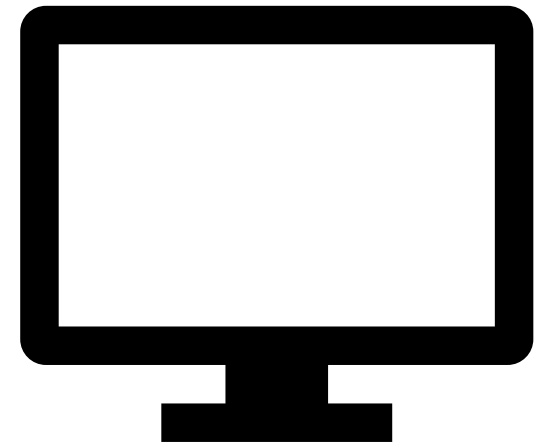
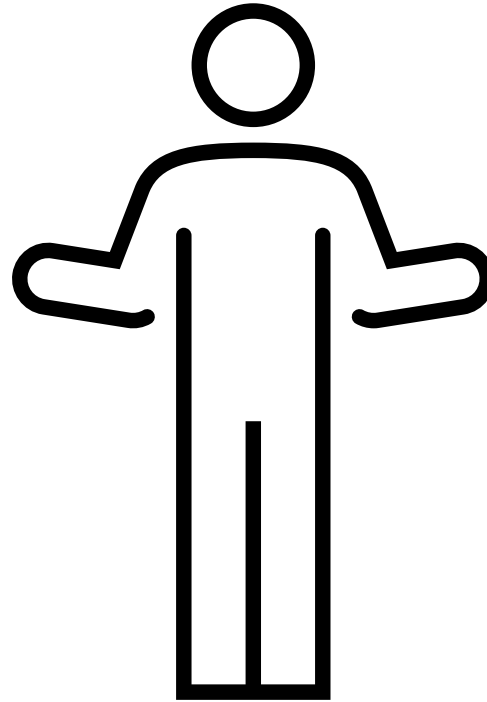
- Name tags!
 - Please make a name tag if you have not already
 - **Write BIG** so I can see your name
 - I will collect your nametags at the end of class, this is how attendance will be taken

About this course

Communicating with data

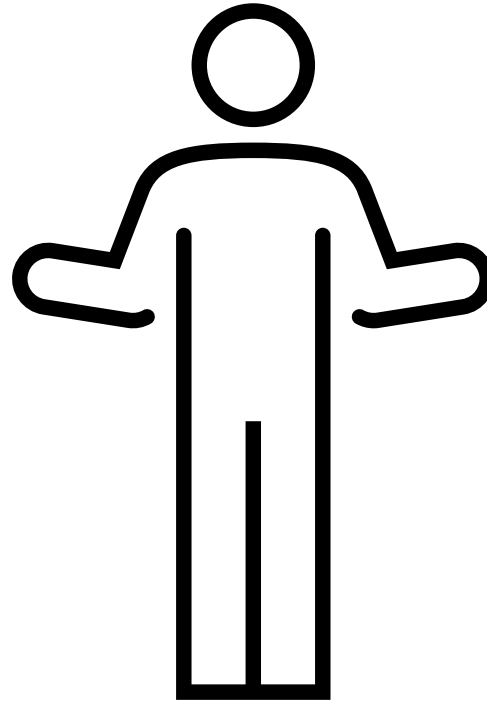
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Communicating with data

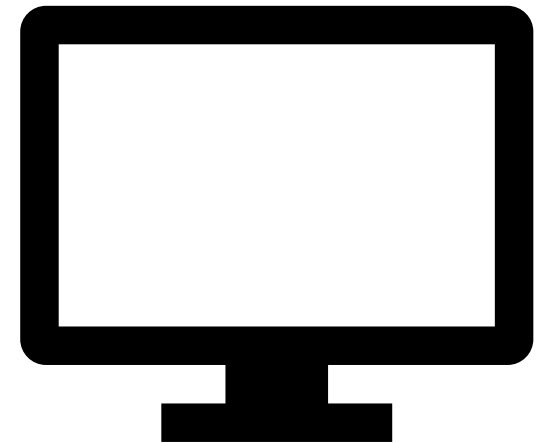


About this course

Communicating with data



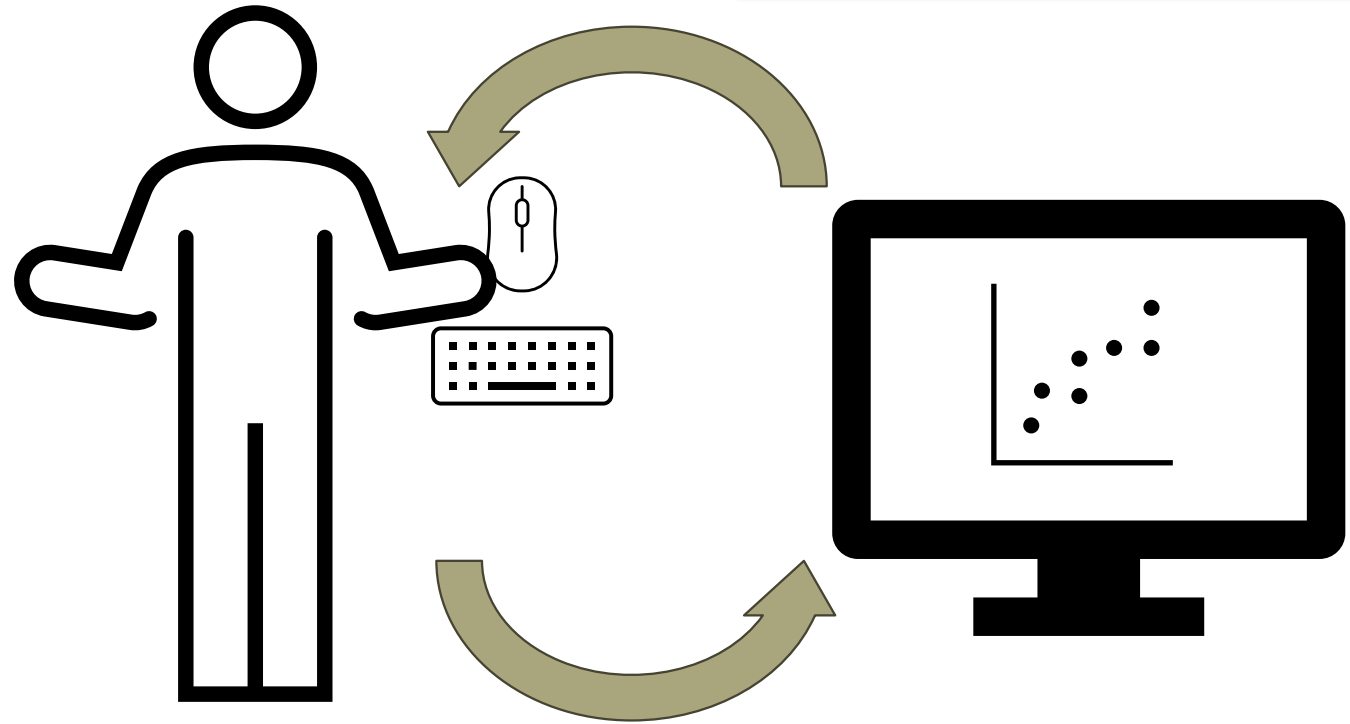
What are the strengths of each wrt data and communication?



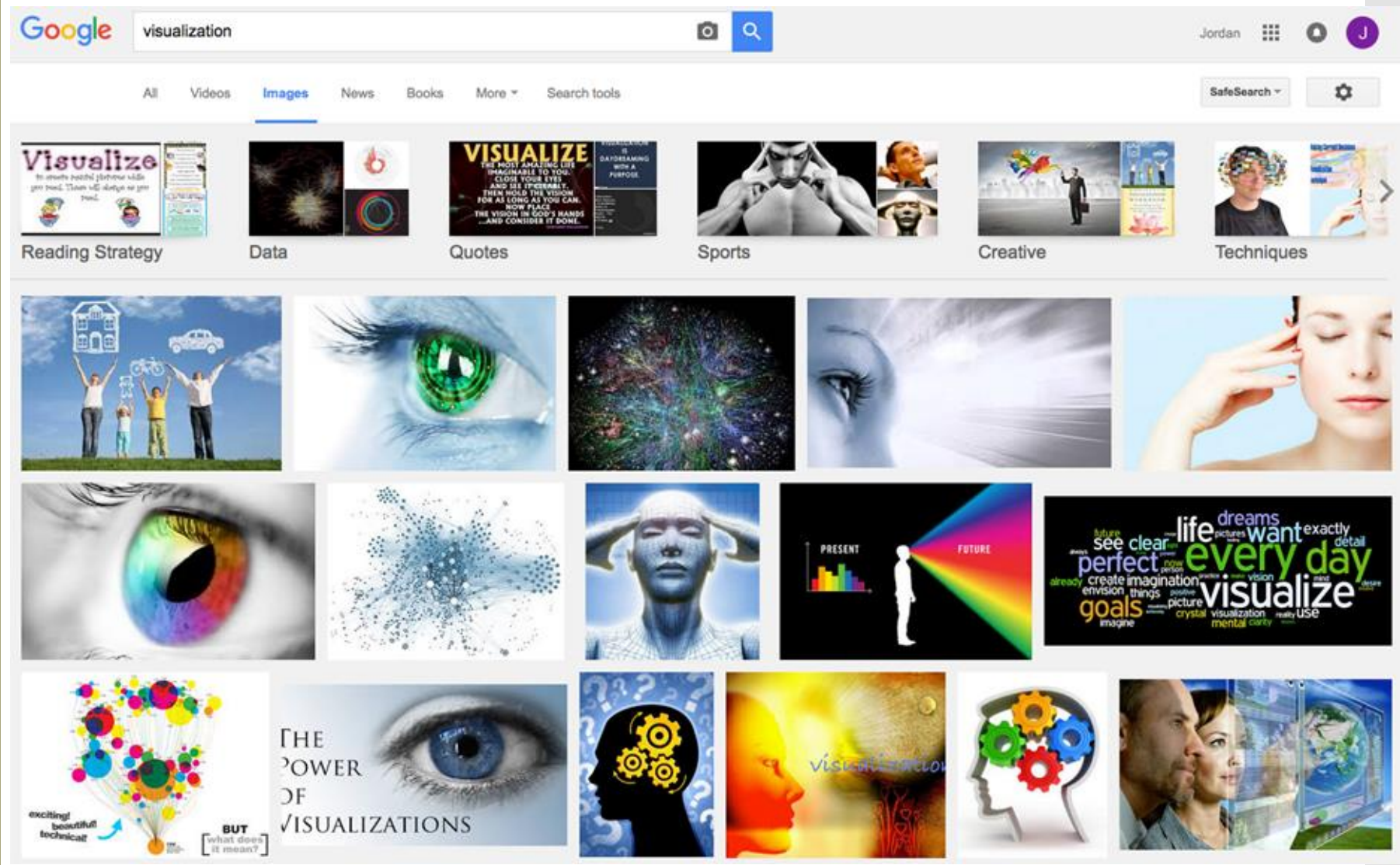
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Communicating with data

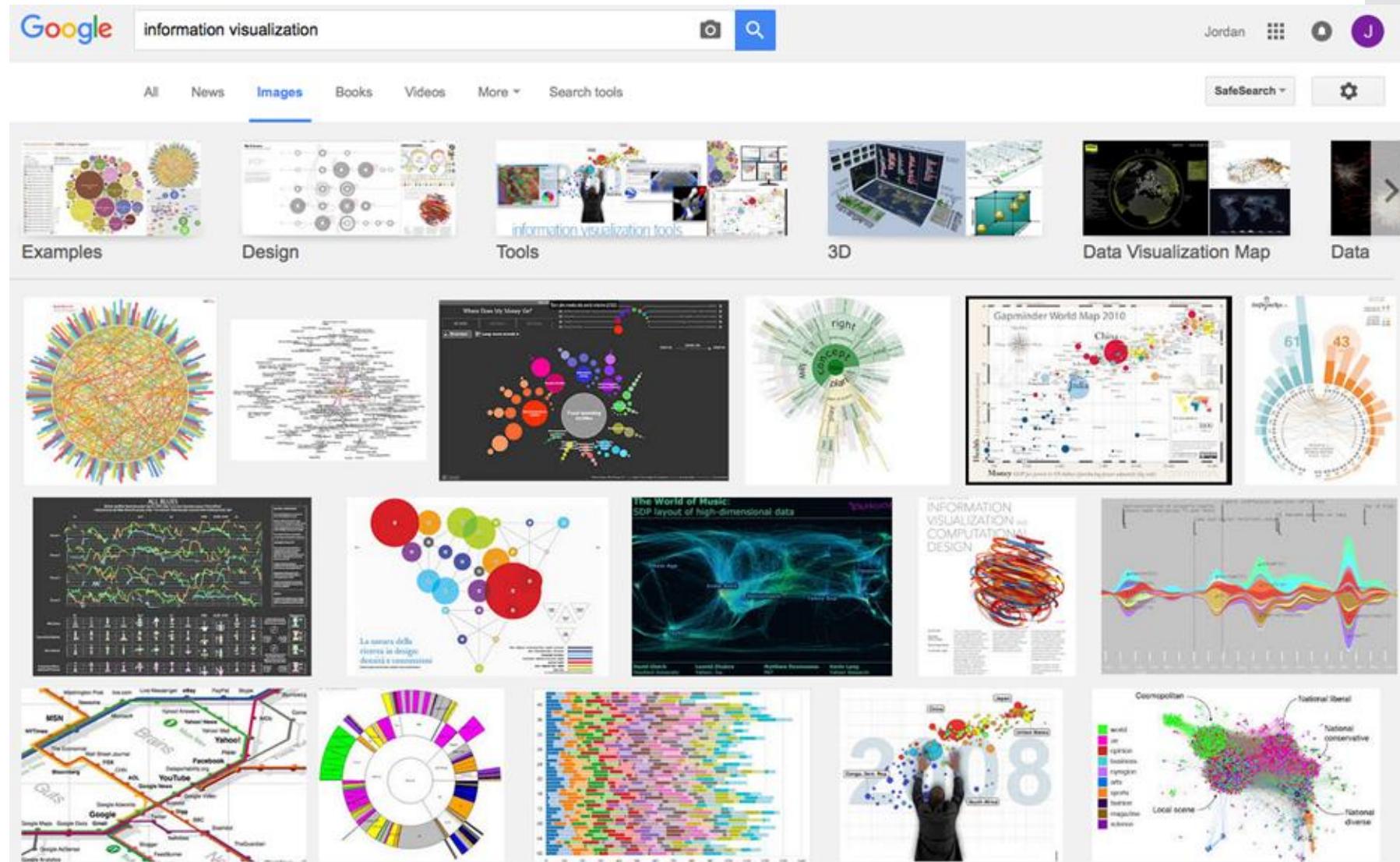
What are the strengths of each wrt data and communication?



What is visualization?



What is visualization?



Perhaps a
more helpful
question:

What are some ways
a “visualization” can be **useful**?

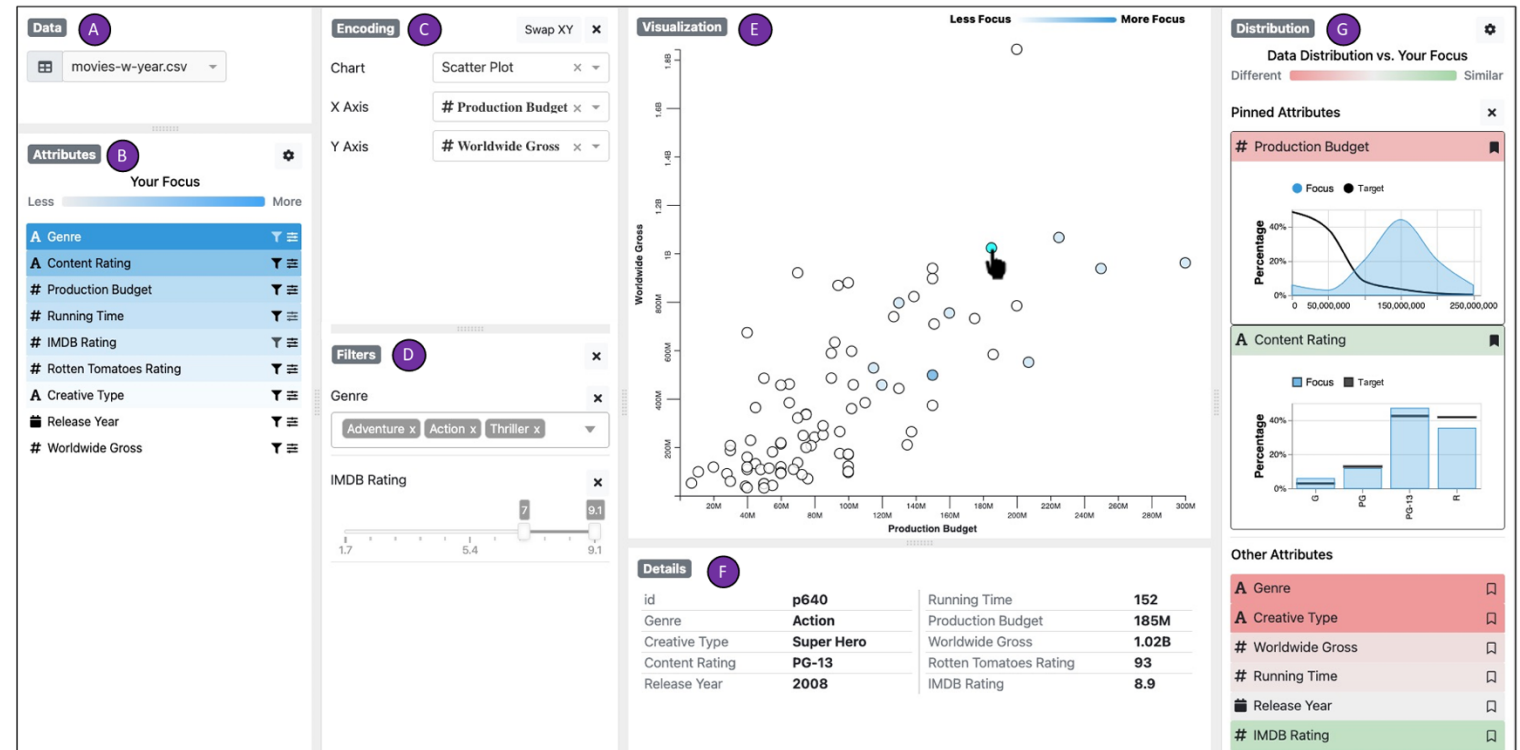
Does it help
you spot
trends?



More info here: http://en.wikipedia.org/wiki/1854_Broad_Street_cholera_outbreak

Lumos: Increasing Awareness of Analytic Behavior during Visual Data Analysis

Arpit Narechania, Adam Coscia, Emily Wall, Alex Endert



Does it help you explore?

<https://lumos-webapp-4aeadb3bf3od.herokuapp.com/>

Does it tell a
story?



Visualization
(def.)

Visual
representations
of data that
reinforce human
cognition



Wait... what is
“data”?



Data: a definition

Data is a set of *variables* that capture various aspects of the world:



*Tuition rates, enrollment numbers,
public vs. private, etc.*

Data: a definition

A dataset also contains a set of *observations* (also called *records*) over these variables. For example:



tuition = \$46,288, *enrollment* = 2,563,
private, etc.

Data: a definition

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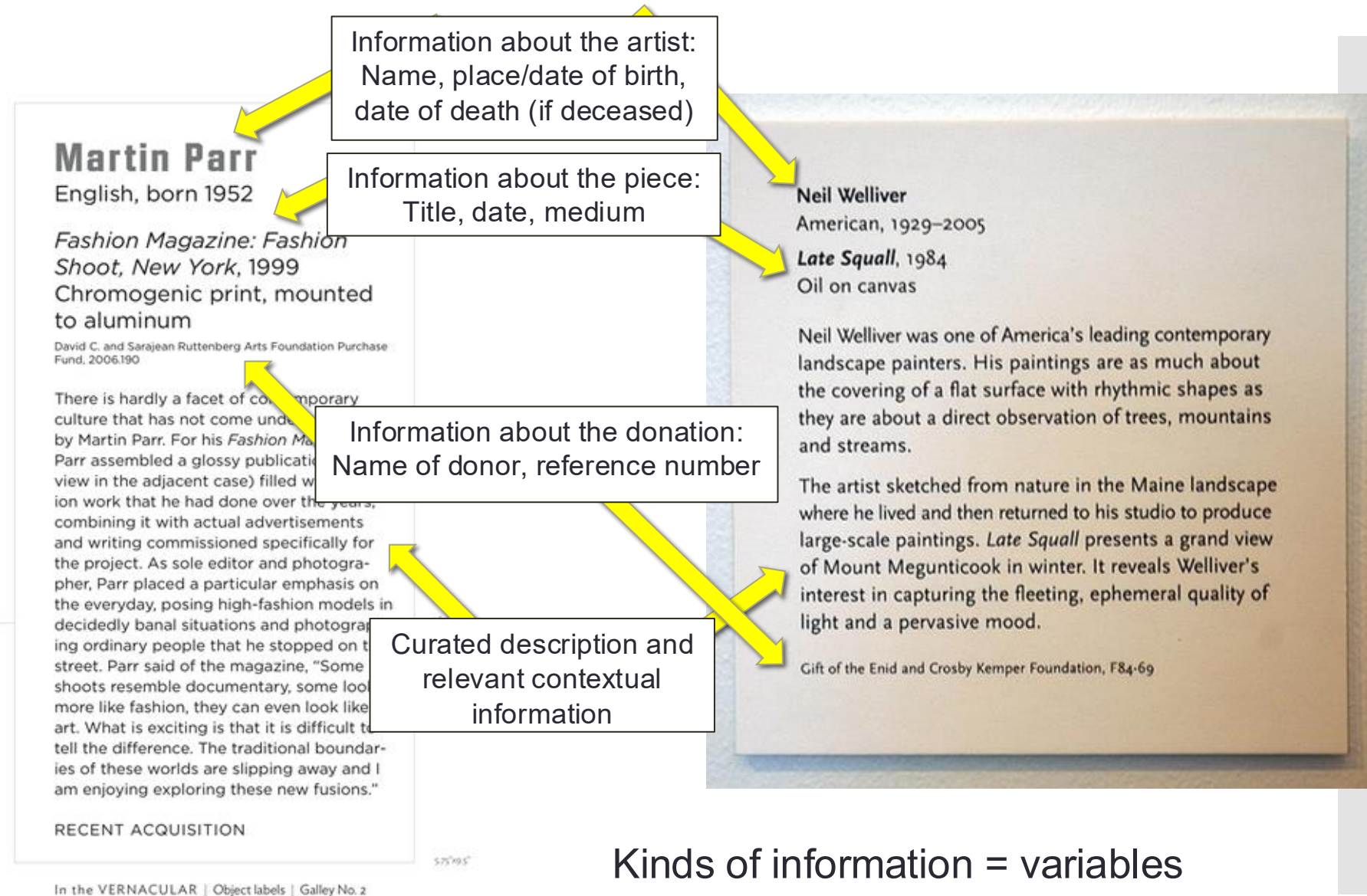
tuition = \$16,115, *enrollment* = 28,635,
public, *etc.*

Also called: Attributes, Dimensions

One way to
think about this:

OBSERVATIONS	VARIABLES			
		Tuition	Enrollment	Public vs. Private
	Smith College	\$46,288	2,563	private
	UMass Amherst	\$16,115	28,635	public
	Hampshire College	\$48,065	1,400	private
	Mount Holyoke College	\$43,886	2,189	private
	Amherst College	\$50,562	1,792	private
	⋮			

Another way to
think about this



Kinds of information = variables
Actual text on the placard = observations

Basic data types

- Nominal
- Ordinal
- Scale / Quantitative
 - Ratio
 - Interval

An **unordered** set $\{...\}$
of non-numeric values

For example:

- Categorical (finite) data
 $\{\text{apple, orange, pear}\}$
 $\{\text{red, green, blue}\}$
- Arbitrary (infinite) data
 $\{\text{"12 Main St. Boston MA", "45 Wall St. New York NY", ...}\}$
 $\{\text{"John Smith", "Jane Doe", ...}\}$

Basic data types

- Nominal
- Ordinal
- Scale / Quantitative
 - Ratio
 - Interval

An **ordered set**
(also known as a tuple)

<...>

For example:

- Numeric: <2, 4, 6, 8>
- Binary: <0, 1>
- Non-numeric:
<G, PG, PG-13, R>

Basic data types

- Nominal
- Ordinal
- Scale / Quantitative
 - Ratio
 - Interval

A **numeric range** [...]

Ratios

- Distance from “absolute zero”
- Can be compared mathematically using division
- For example: height, weight

Intervals

- Ordered numeric elements that can be mathematically manipulated, but cannot be compared as ratios
- E.g.: date, current time

Basic operations

- Nominal (N)
 - Equality: $=$ and $\neq$
 - Frequency: how often does x appear?
- Ordinal (O)
 - Relation to other points: $>$, $<$, $\geq$, $\leq$
 - Distribution: relative frequency
- Quantitative (Q)
 - Other mathematical operations: $(+, -, *, /, \text{etc.})$
 - Descriptive statistics: average, standard deviation, etc.

Why is this important?

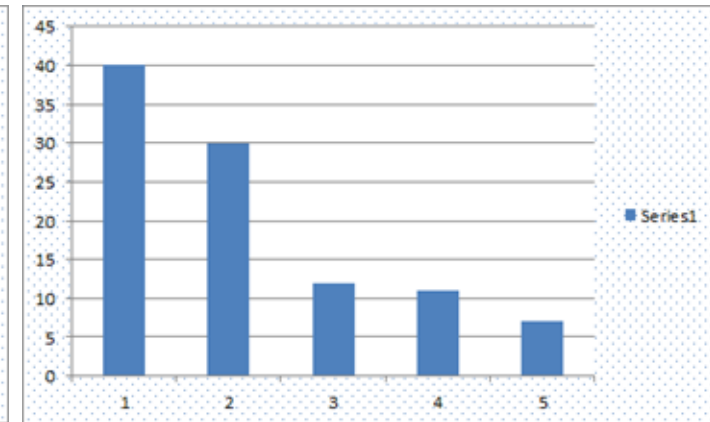
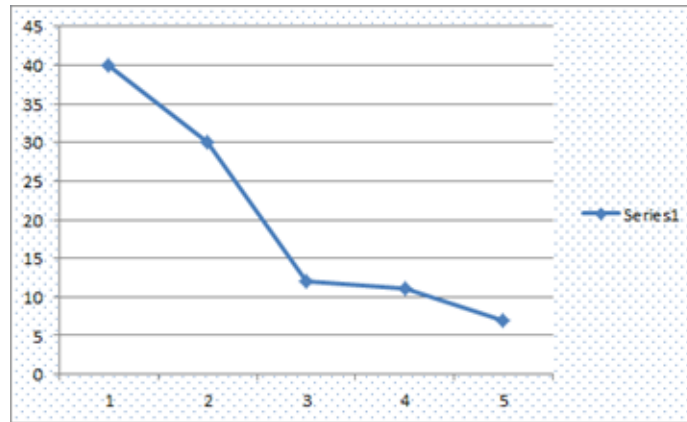
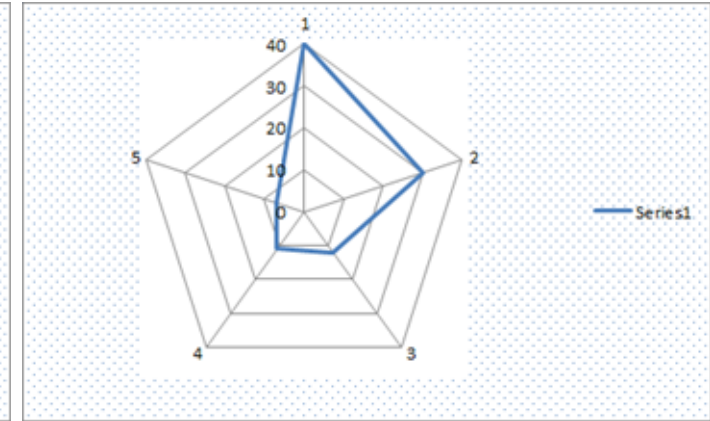
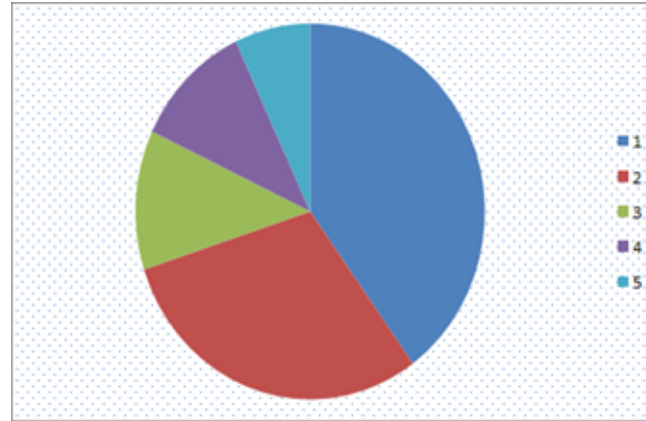
- Data have variables
- Visualizations have variables, too
- To build visualizations, we need to **map** data variables to visual variables

An orange speech bubble with a tail pointing towards the second bullet point.

Like what?

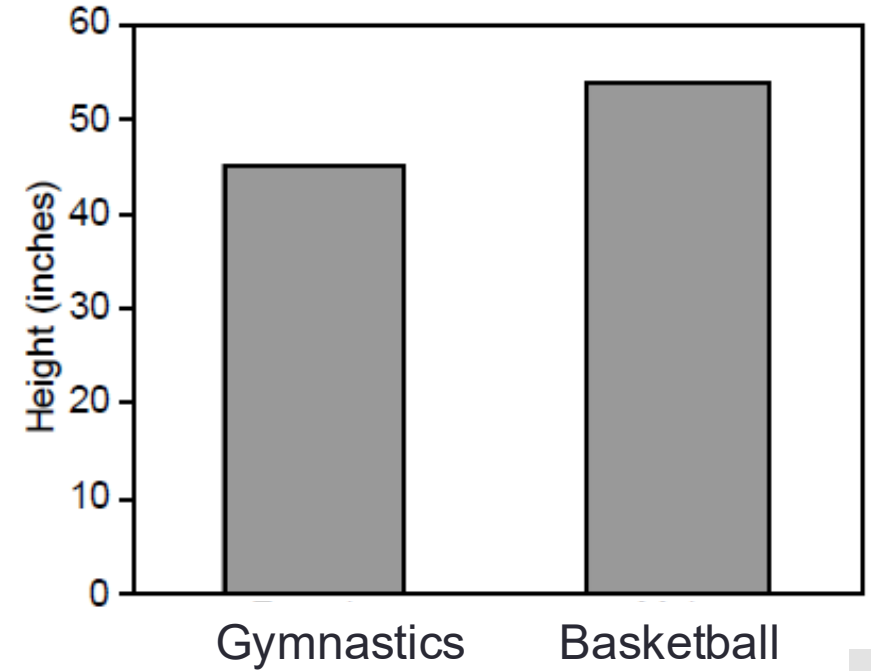
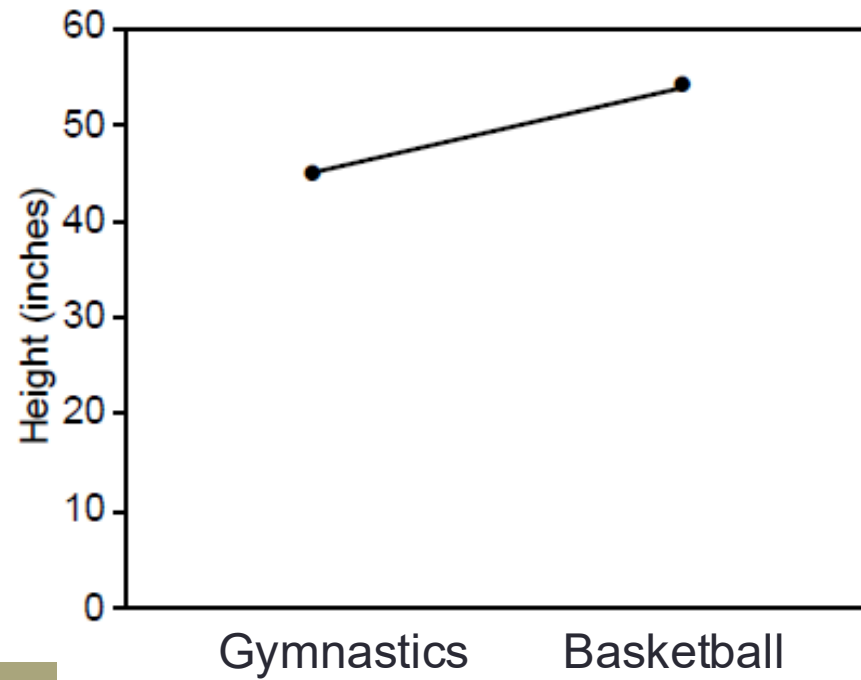
Key question
for this course

Which **data dimension** should be mapped
to which **visual dimension**?



Answer: it depends

Average Height for Youth Sports Participants



What we'll cover in this class

- Data-visual mapping
- Introduction to perception
- Introduction to color
- Embellishment
- Data Storytelling
- Tableau
- Persuasive visualization
- Accessible visualization design
- Visualization literacy
- Data feminism and humanism
- Applications

What we will
not cover in
this class

- Introduction to coding
- Introduction to statistics
- Data wrangling
- Advanced visualization theory (take 235 for this!)

Course Website

- <https://amoscao1.github.io/SDS-CSC109/>
- Syllabus
- Slides
- Schedule
- Assignments
- Links

General information

Time Management:

Smith college guidelines suggest 3 hours of work per credit per week. This is a 4 credit course, so you should expect to spend 12 hours on it per week:

- 2.5 hours in class
- 1-2 hours preparing for class
- **7.5-8.5 hours on homework and outside study**

The purpose of class time is to introduce and explore new topics. To complete assignments, you will often need to look up, learn, and practice material that extends beyond that covered during class meeting times.

Competencies

Core Competencies

1. Map data to appropriate visual channels
2. Accurately encode data values via color
3. Apply perceptual concepts to visualization design
4. Appropriately use visual embellishment
5. Create a data story using storytelling techniques
6. Design and implement an accessible visualization
7. Apply data humanism and feminism ideas to visualizations
8. Write a thoughtful and actionable critique of someone else's work
9. Redesign a visualization based on critiques
10. Positively contribute to the classroom community

Competencies

Each assignment offers an opportunity to demonstrate at least one core competency. Assignments a1, and a2 focus explicitly on data-visual mapping. Assignments a3-6 each require students to produce a visualization (p1), and then assess their work and their peers' work (p2). In these assignments, students may demonstrate competency in as many core competency areas as they wish. However, it is strongly recommended to focus on 2-3 competencies per assignment, so that they can get full attention. Please see the intro slide deck for more detailed information and examples.

Students will receive feedback in class, and via Gradescope on the work they submit.

Competencies

Within each assignment you will see instructions like this:

Competency Demonstration

This is a competency-based course. After you complete your assignment, write a short, succinct, and clear reflection explaining which core competencies you demonstrated in your work and how.

It is recommended that you use this assignment to demonstrate any of the following competencies:

- Map data to appropriate visual channels

A reflection will look something like this:

In this assignment I demonstrated competency in two areas: |

1. Map data to appropriate visual channels
 - a. My bar chart visualizes three variables (1) class year, (2) average number of extracurricular activities, and (3) campus neighborhood. I represented class year via position on the x-axis, which is appropriate as class year is an ordinal variable. I represented average number of extracurricular activities via position on the y-axis, which is appropriate because it is a quantitative variable. And I represented campus neighborhood via hue, which is appropriate because it is a nominal variable. These data visual mappings show that I understand the theory behind choosing appropriate visual channels for different variable types.
2. Create a data story using storytelling techniques
 - a. My scrollytelling visualization is an example of author-driven storytelling. Within the visualization the reader is unable to interact beyond scrolling through the story. As a result, they see the visualizations exactly as I designed them, in the order I chose. I chose this approach because I had a very specific take-away message I wanted the reader to leave with. I thought the best way to convey that message was for the reader to first see visualizations of summary statistics, followed by a deep dive on demographics, and then a visualization representing output of an analysis of demographics and whether they affect financial aid awards.

Competencies

Students will keep a portfolio that demonstrates their competency in each core competency area. It is the students' responsibility to keep their portfolio updated throughout the semester. During finals period, each student will meet with the instructor to review their portfolio and assess competency in each competency area.

Students are invited and encouraged to come to office hours at any point with their in-progress portfolio to discuss progress towards competencies.

Competency-Based Assessment

Competencies

Final Grade	Number of competencies demonstrated fully	Number of competencies demonstrated partially	Number of competencies not attempted
A	10	0	0
A-	9	1	0
B+	9	0	0
B	8	2	0
B-	8	1	0
C+	7	3	1
C	7	2	1
C-	7	1	1
D	6	0	2
E	<6	NA	NA

Late Policy

- All assignments are due as listed on the [Course Schedule](#). Because our lives and learning do not always go as planned, every student will be able grant themselves extensions on homework assignments according to the instructions below.
- Remember, we develop skills through constant practice. This course features weekly assignments, which build from one week to the next, and each assignment is designed to take a full week of work. Delaying the completion of an assignment will put you at a disadvantage going into the next one. Nevertheless, sometimes a little flexibility makes a huge difference, so students are empowered to give themselves extensions when it would aid their ability to demonstrate competency in the material.
- Students who chose to take an extension on a visualization viewing day are still expected to be in class that day to provide feedback to classmates. Moreover, arriving without a visualization to show will negatively impact a student's ability to demonstrate competency 10. When a student takes an extension on p1 of a3-6 it is their responsibility to bring the completed visualization to office hours for instructor feedback and to coordinate a critique session with classmates to complete the assignment.

Revise and Resubmit

You may have heard that this class has unlimited revise and resubmits for all work. Under the competency-based grading model there will no longer be unlimited revise and resubmit. Instead, students have multiple opportunities throughout the semester to demonstrate competency in each of the ten course competencies. In addition, there is a revision opportunity built into the semester via a6.

Comfy Classroom

- Need to stand up and leave for a minutes? Do it.
- Want to sit somewhere other than a chair? Go for it.
- Have a concentration aid? Use it.
- Hungry? Thirsty? Eat and drink (be careful of spills!).
- Have kids and no childcare? Bring your kids.
- Do what you need to do to learn, just be respectful of other learners.

What you'll get

By the end of this course, you will:

- Know how to use visualization to **communicate ideas**
- Know the **foundational methods** and tools available
- Be familiar with **ongoing research** in visualization
- Have (marketable!) experience developing useful visualizations to solve **real problems**

What I expect from you

- You like challenging problems, and you're excited about "figuring stuff out"
- You're willing to get comfortable asking questions
- You're interested in the perspectives of people with very different backgrounds from your own
- You turn things in on time, show up to class, or let me know in advance

What you can expect from me

- I'm flexible w.r.t. the topics we cover:
 - This course is a collaboration
 - If there's something you want to learn that's not on the agenda, speak up!
 - If I'm doing something that doesn't work for you (Font too small on presentations? Speaking too quickly? Using a marker or color you can't see?), please let me know!

I'm here to help you succeed, and I believe you all have the ability to succeed

Questions?